CS228 : Logic for Computer Science

Module 3: Predicate Logic

Lecture 13: Semantics of Predicate Logic

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Logistics

Syllabus for Mid-Semester Exam

- Module 1: Propositional Logic
- Module 2: The Problem of Satisfiability
- All lectures till and including Lecture 11

Mid-Semester Exam

- Venue: LA 201, LA 002; PwD students at KR 125
- Seating arrangement will be shared on Piazza
- 1 A4 size paper allowed as cheatsheet, but MUST be handwritten with blue ink – no photocopy, etc.

Topics Covered till date

Module 1: Propositional Logic

1. Syntax and Semantics ause
2. Formal proof systems
3. Normal Forms

Module 2: The Problem of Satisfiability

1. Propositional Resolution - soundness, termination and completeness
2. The DPLL algorithm
3. Clause learning, the CDCL algorithm, and modern SAT solvers
4. A tool demo of SAT/SMT solvers in practice

Module 3: Predicate Logic

1. Syntax of predicate logic (Lecture 12)
2. Today: semantics of predicate logic

Quick Recap: Syntax of Predicate Logic

- Terms: $t ::= x \mid c \mid f(t, \ldots, t)$
  where x is a variable, c a constant, f a function symbol
- Formulas: $\varphi ::= P(t_1, \ldots, t_n) \mid \neg\varphi \mid \varphi \wedge \varphi \mid \varphi \vee \varphi \mid \varphi \to \varphi \mid \forall x\, \varphi \mid \exists x\, \varphi$
  where P is a predicate symbol
- Free/bound variables: $\mathit{free}(\varphi)$, $\mathit{bnd}(\varphi)$ defined recursively – not complements of each other!
- Substitution: $\varphi[t/x]$ replaces free occurrences of x by t – but only if t is free for x in φ (no variable of t gets captured by a quantifier in φ)!

Today: what do these formulas actually *mean*?

Topic 13.1

Semantics of Predicate Logic

Semantics of FOL

Example 13.1 $\psi_1 := \forall x \forall y (P(x,y) \rightarrow \exists z (\neg(z = x) \wedge \neg(z = y) \wedge P(x,z) \wedge P(z,y)))$

1. What does this mean? English meaning: For every x,y if $P(x,y)$ holds, then we can find z different from x, y such that $P(x,z), P(z,y)$ hold.
2. Is the statement true?
   - Case 1: Suppose,
     - variables take values from real numbers.
     - $P(x,y)$ represents $x < y$
     - Then, ψ_1 is true, i.e., real numbers are dense!
   - Case 2: Instead suppose,
     - variables take values from real numbers, but $P(x,y)$ represents $x \leq y$
     - Then, ψ_1 is false!
   - Case 3: Finally, suppose,
     - variables take values from natural numbers, with $P(x,y)$ representing $x < y$
     - Then, ψ_1 is false, since natural numbers are not dense.

Truth value of any formula φ depends on (i) underlying set from which variables take values and (ii) interpretation of constants, functions and predicates, i.e., Vocabulary!

Semantics of FOL

- Vocabulary $\mathcal{V}$. E.g., $\mathcal{V} = \{a, f, =, P\}$
- $\mathcal{V}$-formula. E.g., $\psi_2 := \exists x(P(x, f(y, a)) \wedge \neg(x = a))$, $var(\psi_2) = \{x, y\}$, $free(\psi_2) = \{y\}$.

Truth of φ depends on

1. Universe U from which variables take values, e.g., $\mathbb{N}$
2. Interpretation of $\mathcal{V}$ on U
   - Each constant to an element of U, e.g., $a \mapsto 0$
   - Each n-ary function symbol to a function from U^n to U, e.g., $f(u, v) = u + v$.
   - Each m-ary predicate symbol to a subset of U^m, e.g.,
     - Interpret for $=$: $\{(c, c) \mid c \in \mathbb{N}\}$ - fixed interpretation!
     - Interpret for P : $\{(c, d) \mid c, d \in U, c < d\}$

   This defines a $\mathcal{V}$-structure (or Model) $M = (U^M, (a^M, f^M, P^M))$, e.g., $(\mathbb{N}, (0, +, <))$
3. We also need an environment $\alpha : var(\varphi) \to U$, e.g., $\alpha(y) = 2, \alpha(x) = 1$.
   - We will also use $\alpha[x \mapsto u]$ to denote the env which maps x to u (other vars are mapped according to α).

Semantics of FOL: An Example First

- **Example 13.2** Consider again $\psi_2 := \exists x(P(x, f(y, a)) \wedge \neg(x = a))$ over Vocabulary $\mathcal{V} = \{a, f, =, P\}$
- Also, let $M = (\mathbb{N}, (0, +, <))$ be a $\mathcal{V}$-structure, and $\alpha(x) = 0, \alpha(y) = 2$ be the env.
- Does M, α satisfy ψ_2?
  - First compute $f(y, a) = y + a = 2 + 0 = 2$ (using $\alpha(y) = 2, a^M = 0$).
  - Is $P(x, f(y, a))$ true, i.e. is $x < 2$?
  - Is $\neg(x = a)$ true, i.e. is $x \neq 0$?
  - Is their conjunction true, i.e. is $x < 2$ and $x \neq 0$?
  - Finally, is there a value of x from the universe $\mathbb{N}$ that makes their conjunction true?
    - With $x = 0$ it is not satisfied.
    - But, $x = 1$ works! And that is all we need.
    - In other words, the env α, where we set $x = 1$ (ignoring the earlier value) satisfies the formula, since $(1 < 2 \wedge 1 \neq 0)$ is true!
  - Thus, we can conclude $M, \alpha \models \psi_2$!

Now, let's make this definition precise and generic!

Semantics of FOL: Formal Definition

Given $\mathcal{V}$-structure/model M and env α, we define $M, \alpha \models \varphi$ (satisfaction relation)

By structural induction! First we extend α to $\overline{\alpha} : \mathit{Terms}(\varphi) \to U^M$, by

- If t is a variable x, then $\overline{\alpha}(t) = \alpha(x)$
- If t is $f(t_1, \ldots, t_m)$, then $\overline{\alpha}(t) = f^M(\overline{\alpha}(t_1), \ldots, \overline{\alpha}(t_m))$
  In Example 13.2, we have $\overline{\alpha}(f(y, a)) = f^M(\alpha(y), a^M) = 2 + 0 = 2$

- $M, \alpha \models P(t_1, \ldots t_m)$ iff $(\overline{\alpha}(t_1), \ldots \overline{\alpha}(t_m)) \in P^M$ (atomic formulas)
- $M, \alpha \models \neg\varphi$ iff $M, \alpha \not\models \varphi$.
- $M, \alpha \models \varphi_1 \wedge \varphi_2$ iff $M, \alpha \models \varphi_1$ and $M, \alpha \models \varphi_2$.
- $M, \alpha \models \exists x\varphi$ iff there exists some $u \in U^M$ s.t., $M, \alpha[x \mapsto u] \models \varphi$.
- $M, \alpha \models \forall x\varphi$ iff for all $u \in U^M$, $M, \alpha[x \mapsto u] \models \varphi$.

We omit $\vee, \to, \Leftrightarrow$ here since they reduce to $\wedge, \neg$ in the usual way (e.g. $\varphi_1 \to \varphi_2 \equiv \neg\varphi_1 \vee \varphi_2$) – same treatment as $\wedge$ above. **Exercise 13.1** In the e.g., what if we chose (i) $\alpha(y) = 0$? (ii) $\alpha(x) = 5$? (iii) $\alpha(x) = 1$?

Satisfaction only depends on the "free" environment

Exercise 13.2

Show that if α, α' are identical on the set of free variables, then for any formula φ and any $\mathcal{V}$-structure M, $M, \alpha \models \varphi$ iff $M, \alpha' \models \varphi$.

hint: use structural induction!

- Thus, if φ has no free variables, i.e., it is a sentence, then α does not matter!

For a sentence φ, we just write $M \models \varphi$.

Recall: Truth Depends on the Structure

In Example 13.1 $\psi_1 := \forall x \forall y (P(x,y) \to \exists z (\neg(z = x) \wedge \neg(z = y) \wedge P(x,z) \wedge P(z,y)))$

- For structure $M_1 = (\mathbb{R}, <)$: $M_1 \models \psi_1$ (density!)
- For structure $M_2 = (\mathbb{R}, \leq)$: $M_2 \not\models \psi_1$
- For structure $M_3 = (\mathbb{N}, <)$: $M_3 \not\models \psi_1$

What is the vocab here?
Why is there no α?

In Example 13.2 $\psi_2 := \exists x (P(x, f(y, a)) \wedge \neg(x = a))$ over $\mathcal{V} = \{a, f, =, P\}$

- For $M_4 = (\mathbb{N}, (0, +, <))$ with $\alpha(x) = 0, \alpha(y) = 2$: $M_4, \alpha \models \psi_2$ (witness $x = 1$)
- For $M_4 = (\mathbb{N}, (0, +, <))$ with $\alpha'(x) = 0, \alpha'(y) = 0$: $M_4, \alpha' \not\models \psi_2$ (need $x < 0$, no natural number works!)

Exercise 13.3 Find a *different* structure M' (and/or environment α'') for formula ψ_2, and determine whether $M', \alpha'' \models \psi_2$.
What happens when $M' = (\mathbb{N}, (1, \times, <))$? Is there an env which makes the formula true/false?

Revisiting substitution

Exercise 13.4 Let $\psi_3(x) := \exists y\,(y > x)$ over $\mathcal{V} = \{>\}$, with $M = (\mathbb{N}, >)$.

1. What is *free*(ψ_3)? For any α, does $M, \alpha \models \psi_3$ hold?
2. Is $t = y$ free for x in ψ_3? If we substitute anyway, what is $\psi_3[y/x]$? What does this tell us about the substitution definition?

free$(\psi_3) = \{x\}$. For *any* α, is there $u \in \mathbb{N}$ with $u > \alpha(x)$? Yes, always! Numbers are unbounded above, so $M, \alpha \models \psi_3$ for every α.

Recall Substitution in FOL: $t = y$ is not free for x in ψ_3: (y occurs in t, and the free x is under the scope of $\exists y$)

- Substituting anyway, step by step:

$$\exists y\,(y > x) \quad \longrightarrow \quad \exists y\,(y > y)$$

- So $\psi_3[y/x] = \exists y\,(y > y)$
- Is *this* true? No – always false! No number is bigger than itself.

The truth value flipped. This is exactly why substitution is restricted to terms that are free for x.

Semantic entailment

Once we fix the vocab $\mathcal{V}$, we just write structure instead of $\mathcal{V}$-structure.

Let Γ be a (possibly infinite) set of formulas and let ψ be a formula in predicate logic.

- **Semantic entailment** $\Gamma \models \psi$ holds iff for all M and α, whenever $M, \alpha \models \varphi$ holds for all $\varphi \in \Gamma$, then $M, \alpha \models \psi$ holds as well.
- ψ is **satisfiable** iff there is some M and α such that $M, \alpha \models \psi$ holds.
- ψ is **valid** iff $M, \alpha \models \psi$ holds for all M and α.
- Γ is **consistent** iff there is at least one M and α such that $M, \alpha \models \varphi$ for all $\varphi \in \Gamma$.

Note: M ranges over *all* universes and *all* interpretations of $\mathcal{V}$ on them – not a fixed universe!

Exercise 13.5 Give 2 examples of valid and 2 of satisfiable (but not valid) formulas with the vocab given in Example 13.2.

Exercise 13.5 Solution

Vocab: constant a, unary function f, binary predicate P (plus always-present $=$).

Valid (true for every M and α, regardless of interpretation)

1. $a = a$ – equality has a *fixed* interpretation (always literal identity), so this holds no matter how a is interpreted.
2. $\forall x\,(P(x, f(a)) \to P(x, f(a)))$ – an instance of the propositional tautology $\varphi \to \varphi$, lifted into predicate logic; true regardless of how P, f, a are interpreted.

Satisfiable but *not* valid

1. $P(a, a)$ – true if we pick M with $P^M = U \times U$ (P always holds); false in our familiar $M_4 = (\mathbb{N}, (0, +, <))$ since $0 < 0$ is false. Satisfiable, not valid.
2. $\exists x\, P(x, a)$ – true if $U = \mathbb{Z}$, P interpreted as "$<$", $a = 0$ (witness $x = -1$); false in $M_4 = (\mathbb{N}, (0, +, <))$ since no natural number is < 0. Satisfiable, not valid.

Semantic equivalences

Definition

$\varphi \equiv \psi$ iff $\{\varphi\} \models \psi$ and $\{\psi\} \models \varphi$.

Exercise 13.6 Prove the following: Quantifier equivalences

- $\forall\, x\varphi \equiv \neg\exists\, x\neg\varphi$
- $\forall x \forall y \varphi \equiv \forall y \forall x \varphi$
- $\forall x(\varphi_1 \wedge \varphi_2) \equiv (\forall x \varphi_1) \wedge (\forall x \varphi_2)$
- $\exists x(\varphi_1 \vee \varphi_2) \equiv (\exists x \varphi_1) \vee (\exists x \varphi_2)$

Validity and Satisfiability for predicate logic

- For propositional logic, check validity just required to write down the truth table.
- Can we do something similar for predicate logic, in particular FOL?

No! Checking validity is undecidable!

Undecidable means: there is *no algorithm* which, given any formula, always correctly answers "valid" or "not valid" in finite time.

- In fact this is the case for a vocabulary with 1 constant, 2 unary functions and 1 binary predicate! (Church, 1936)
- Encoding of Post's correspondence problem.



What about satisfiability? Also undecidable!